

Multi-omics fusion based on attention mechanism for survival and drug response prediction in Digestive System Tumors

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ABSTRACT

In recent decades, digestive system tumors (DST) have become the primary cause of cancer-related deaths worldwide. Improving tumor prognosis and drug response prediction holds significant importance in personalized medicine. In order to construct an effective and interpretable model that integrates multiple omics data, the **Multi-omics Fusion Graph Attention Network (MFGAN)** is proposed for survival prediction and drug response prediction in DST. In this method, the similarity matrix of each kind of omics data is calculated and a new graph structure is learned by Graph Transformer (GT). Next, to learn the features of every kind of omics data from the TCGA and GDSC databases, the Graph Attention Network (GAN) is used. Lastly, the View Correlation Discovery Network (VCDN) combines different types of omics data features to predict survival risk and drug response. Among them, the survival prediction results showed an improvement of up to 9% relative to other methods in terms of the c-index metric, with a minimum improvement of 2.6%. In terms of drug response prediction performance, there was an improvement of 4%. The ablation experiment demonstrates MFGAN's feature integration capability, and the functional enrichment analysis for significant genes explains the functional characteristics of the model. Furthermore, the prediction of unknown tumor drug response demonstrates the model's prediction ability. The proposed high-prediction comprehensive model could have important potential value for the personalized medicine of DST.

1. Introduction

Cancer accounts for the highest mortality rate worldwide, with digestive system tumors (DST) being one of the most prevalent forms, including gastric, liver, esophageal, colorectal, and pancreatic cancer. According to World Health Organization (WHO), around 30,000 individuals die annually due to DST, placing it fourth among the top reasons for worldwide death [1]. DST has perpetuated as a significant health burden worldwide, representing 26% of all cancer detection and 35% of the whole cancer-associated mortalities in 2018. Colorectal and gastric cancers are highly prevalent forms of malignant diseases, which have been associated with high mortality rates worldwide. Specifically, colorectal cancer has emerged as the second most lethal type of cancer, while gastric cancer ranks the third highest number of cancer

fatalities globally [2]. Recent data for individuals diagnosed with DST between 2012 and 2018 have shown a low 5-year survival rate of only 36%, 47%, and 44% for liver cancer, esophageal cancer, and pancreatic cancer in cases with no tumor spread. For those where the tumor had metastasized to nearby areas or lymph nodes, survival rates decreased sharply to 15%, 13%, 33%, and 26% for pancreatic, liver, gastric, and esophageal cancers, respectively [3]. Though various prognostic markers exist, the high mortality rate and poor prognosis of DST highlight the need for effective and robust prognostication systems to support personalized treatment and achieve better outcomes with fewer side effects [4,5]. Uncovering novel therapeutic agents remains a major priority in cancer research and clinical practice. The development of new drugs for cancer treatment and other diseases has

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traditionally relied on animal testing and laboratory screening methods. Despite their effectiveness in identifying promising drug candidates, these methods suffer from several limitations, including high costs and resource-intensive procedures [6]. For this reason, predicting each patient's response to medication based on their genomic features holds vital clinical significance.

Recent advancements in high-throughput sequencing has allowed for analyzing various genomic covariates related to cancer patient survival, such as DNA mutations, gene expression, and copy number variations (CNVs), to predict patient survival rates [7]. The Cox proportional hazards (PH) model is the most widely used method to predict patient survival and is formulated as a regression problem. This semi-parametric model employs covariates to estimate the relative risk, also known as the hazard ratio, in patients [8]. This model's simplicity may not lend itself well to certain applications, such as generating personalized treatment recommendations. Faraggi and Simon introduced the first artificial neural network (ANN) model to predict patient survival rates, which extended the Cox proportional hazards model in a nonlinear manner. The model developed by the researchers used four clinical input to replicate the survival of individuals with prostate cancer. However, due to its simplistic nature, their model was unsuitable for handling large-scale input data [9]. Deep learning models have shown remarkable efficacy in diverse domains, such as autonomous feature extraction and survival prediction, in recent years. Yang et al. integrated deep neural networks (DNNs) with the Cox model to develop a prognostic score estimator for grouping patients according to common clinical and pathological factors [10]. Katzman et al. proposed the deepsurv model, which is a methodology using deep learning networks that can predict the impact of covariates on hazard rates and provide personalized treatment recommendations to patients using baseline data as input [11]. Combining different types of data sources has been demonstrated to improve survival model accuracy. Using mRNA, CNV data, and gene expression, Arya and Saha developed a unique model called SiGaAtCNN to categorize patients by their survival duration [12]. Zhao et al. presented a deep learning framework called DeepOmics, which can grouping genes by their functional properties by multiple omics data and improve survival prediction by providing a scalable and flexible solution [13].

Due to the heterogeneity of cancer, treatment effects can vary among patients with the same cancer type who receive identical treatment. Predicting cell drug response (CDR) can aid in developing personalized treatment plans [14]. With the increasing availability and diversity of genomic and transcriptomic data, computational approaches have been established as a promising technique for predicting drug sensitivity or resistance in cancer cells. Large-scale pharmacogenomic databases, such as the Cancer Cell Line Encyclopedia (CCLE) [15] and the Genomics of Drug Sensitivity in Cancer (GDSC) [16], provide an extensive array of omics data from different types of cell lines, encompassing various molecular features and drug response profiles. Thus, these databases have become an essential resource for developing novel machine learning-based approaches, including support vector machines (SVM) [17], Bayesian multitask multiple kernel learning [18], random forests (RFs) [19], and neural network models [20]. These methods allow for the integration of multiple sources of genomic and transcriptomic information to predict drug response. Overall, these datasets and predictive models offer tremendous potential for advancing precision medicine and improving patient outcomes in cancer research. However, genomic and cell line omics data are often represented in high-dimensional form, and the sample size is small, indicating significant potential for improvement in prediction performance and model generalization [21,22]. Deep learning methods are particularly well-suited for predicting drug responses utilizing omics data from both cell lines and pharmacology due to their ability to effectively handle large, high-dimensional datasets with noise. In contrast to other machine learning algorithms, deep learning approaches offer a more powerful tool for extracting complex patterns and relationships from these types

of data [23,24]. CDRscan is composed of five convolutional neural networks (CNNs), with four adopting a “dual convergence structure”. The dual convergence model isolates input data types for convolutional operations. The resulting convolutional feature sets are then merged and convolved again to predict the IC50 values for cell line-drug pairs from GDSC. Nonetheless, relying solely on genomic mutations as a representation of cancer cell heterogeneity is inadequate for predicting drug response [25]. Another model, tCNN, uses two convolutional networks to learn the one-hot encoding representation of drug SMILES strings and the representation of genomic mutation data, respectively. After creating these representations, they are incorporated into the final prediction network, which can be further optimized for improved performance [26]. MOLI is a method for integrating multiple types of omics data approach that leverages mutations, copy number variations, and gene expression data. To optimize the model, MOLI employs a combined loss function that includes both binary cross-entropy and ternary loss [27]. In contrast, DeepCDR utilizes the Unified Graph Convolutional Network (UGCN) to capture the drug representation and integrate multiple omics data for more accurate IC50 prediction. However, this model requires improvement in terms of density regarding its use of mutation features [28].

In the realm of deep learning, there exists a subfield that deals specifically with non-Euclidean spaces, GAN have produced remarkable results in multiple applications utilizing graph-structured data. GAN uses the graph Laplacian principle to convolve input graphs, instead of fixed grid convolutions used for one-dimensional and two-dimensional Euclidean structural data. It has been widely used for social network and physical system research [29,30]. As a result, GAN models have recently been applied to construct multi-omics data analysis frameworks for classifying biomedical applications. Given the exponential growth of biological network data, GAN have emerged as a vital tool for survival and drug response prediction [31–34]. In comparison, GAN achieves better performance by utilizing the similarity network to describe the correlation between omics features and samples. Therefore, we aim to use multi-omics GAN models to predict survival and drug response of DST. In summary, there are challenges in the lack of interpretability and effective integration of multi-omics data in deep learning methods for survival prediction and drug response prediction. This study proposes a new multiomics integration model, MFGAN, for survival prediction and drug response prediction in DST (see Fig. 1). MFGAN takes gene expression, DNA methylation, copy number variation, and other omic data as input. After feature selection, sample similarity matrices are calculated for each omics data type. To uncover latent correlations between samples, the GT module is used to derive a structural matrix from the similarity matrix of each omics data type. The multiple structural matrices are then combined to form a new graph structure. Next, the GAN method is used to integrate the omics data and the new graph structure to learn features. **Finally, VCDN [35] is used to explore the cross-omics correlation between the features extracted by GAN, enabling efficient multiomics integration for survival prediction and drug response prediction.** The high performance achieved by MFGAN demonstrates the model's usefulness and versatility. Omics data ablation experiments illustrate the need for using various omics data types, and functional enrichment analysis of critical genes demonstrates that MFGAN can identify biomarkers related to prognosis and drug response. In addition, our study shows that MFGAN can be used to make accurate predictions for unknown drug response scenarios. Our proposed integrative multi-omics prediction model has the potential to aid clinicians in making personalized and clinically informed decisions regarding patient treatment strategies.

2. Materials and methods

2.1. Data for survival prediction and preprocessing

We analyzed three types of omics data obtained from The Cancer Genome Atlas (TCGA) for six different types of cancer: colon adenocarcinoma (COAD), esophageal squamous cell carcinoma (ESCA), liver

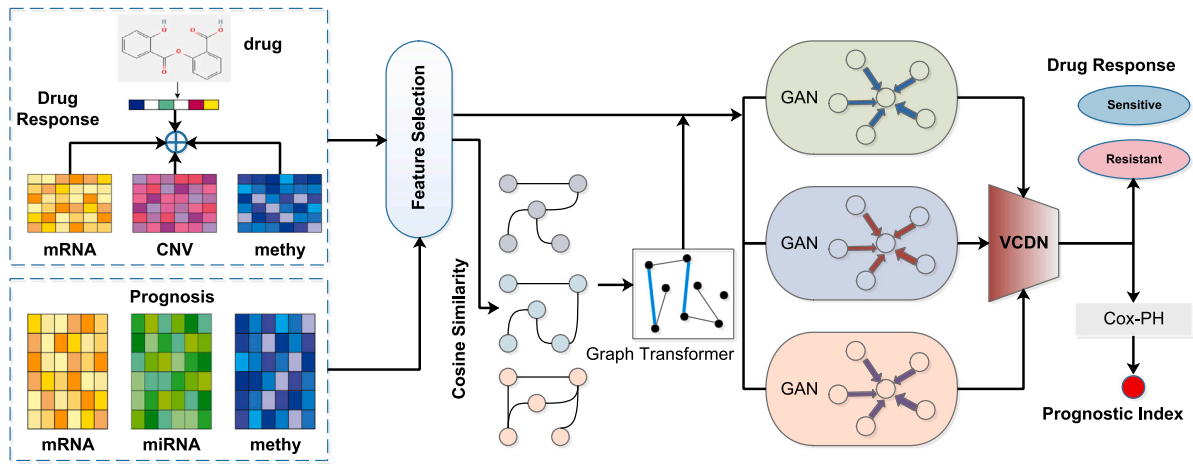


Fig. 1. Diagram of MFGAN structure. The mRNA, methylation, and miRNA serve as inputs for predicting survival risk, and the resulting output is a risk coefficient that reflects the patient's relative risk. When predicting drug response, mRNA, methylation, CNV, and drug data are utilized as inputs, and the output is either resistance or sensitivity.

Table 1
Distribution of samples and features.

	Event = 0	Event = 1	Total	mRNA	methy	miRNA
COAD	226	62	288	444	230	69
ESCA	96	56	152	183	71	99
LIHC	242	122	364	1232	249	234
READ	80	14	94	187	139	50
PAAD	89	89	178	848	280	209
STAD	215	112	327	433	273	46

hepatocellular carcinoma (LIHC), rectum adenocarcinoma (READ), stomach adenocarcinoma (STAD), and pancreatic adenocarcinoma (PAAD). Specifically, we utilized RNA-Seq data, miRNA sequencing data, and DNA methylation data, along with comprehensive information on overall survival status and duration [36]. We removed patients with overall survival duration below 30 days, a total of 1403 samples remained, including 455 cases of death and 948 cases of survival. To eliminate irrelevant noise and avoid false experimental interference, appropriate preprocessing of the genomic data was performed to ensure clear results. Firstly, features with missing values of 80% or more were discarded. In our study, we first imputed missing values in the dataset using the “impute” function in the R software package. Next, we further filtered the data by removing features that do not contain any signal (mean equals to zero) or low variance. As various forms of omics data have varying ranges, we applied different variance filtering thresholds to each type, setting a value of 0.1 was used for mRNA expression data, while a value of 0.001 was utilized for DNA methylation data. Due to the limited amount of available features in miRNA expression data, we only retained features with constant variance (zero variance). After this preprocessing stage, each type of data was scaled individually to values between 0 and 1 using a linear transformation method. Table 1 provides comprehensive details of the preprocessed data used for the training of MFGAN.

2.2. Data for drug response and preprocessing

For our study, we drew drug response data from two databases-GDSC [37] and CCLE [15]. The GDSC database supplied IC50 values, with each value denoting a drug's interaction with a cancer cell line in large-scale drug screening, while the CCLE database provided access to genomic copy number variation, gene expression and DNA methylation data for more than 1000 cancer cell lines. To establish a baseline for measuring CDR, we downloaded hundreds of IC50 values (that were natural logarithm transformed) from the GDSC database and excluded

those lacking PubChem ID with missing omics data for cancer cell lines. We selected cell lines relevant to DST and annotated each based on its tumor type in accordance with the TCGA study. In total, we collected data from 113 cancer cell lines and 153 drugs encompassing 13,774 known cell line-drug pairs. However, out of all $113 \times 153 = 17,289$ cell line-drug pairs, we found that approximately 20.5% (3515 pairs) of IC50 values were missing. We classified the CDR into two groups-sensitive and resistant groups based on the IC50 threshold of the drug. Regarding gene expression data, each cell line was denoted as a feature vector consisting of 697 dimensions, while DNA methylation data was obtained from processed bisulfite sequencing data within the promoter's 1 kb upstream TSS region. We addressed missing values by employing median imputation. Lastly, copy number variation data was represented as a 710-dimensional feature vector for each cell line, while DNA methylation information encompassed an 808-dimensional feature vector.

2.3. Model performance evaluation metrics

The concordance index is a widely used prognostic evaluation indicator in this study, ranging from 0 to 1 [38]. It serves as an evaluation tool for the performance of prognostic models [39] and is analogous to the area under the ROC curve (AUC) [40], which assesses the model's ability to discriminate between surviving and deceased populations. A concordance index (C-index) value of 0.5 implies that the result or action being evaluated has no significant impact or improvement compared to making a random choice or taking a random action, while a higher value above 0.5 denotes superior performance of the prognostic model. AUC and AUPR are two metrics utilized to assess drug response models. All experiments were conducted using cross-validation [41]. A 5-fold cross-validation (5-CV) is applied to the cross-validation set by randomly partitioning responses into five equal segments. The hyperparameters of MFGAN are determined based on the 5-CV results and are subsequently utilized for other experiments.

2.4. Survival analysis

In order to assess the results of MFGAN on 5-fold cross-validation testing sets, survival analysis was conducted using the log-rank test [42]. The Kaplan–Meier survival curves were made by dividing all patients into groups with low and high levels of risk based on their median hazard ratio. The log-rank p -value corresponding to the ability of the model to distinguish between these two risk groups indicates the effectiveness of the model, with lower p -values demonstrating superior performance.

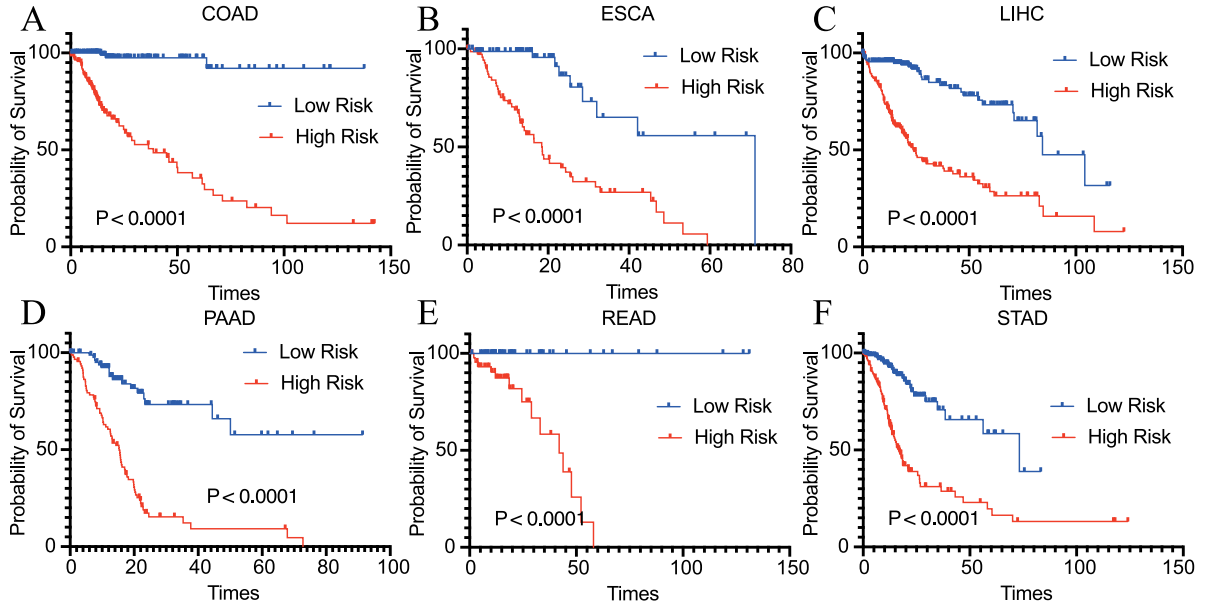


Fig. 2. Based on the risk coefficient's median value, the specimens were divided into groups of high and low risk, and their KM survival curves were plotted. (A) COAD, (B) ESCA, (C) LIHC, (D) PAAD, (E) READ, and (F) STAD.

2.5. Adjacency matrix

The adjacency matrix A necessary for GAN is calculated through cosine similarity between samples. In order to establish a reliable adjacency matrix, a threshold value, denoted as w and set as a hyperparameter, is used. To be precise, if the value of cosine similarity between two given samples exceeds the threshold w , then it is represented in the corresponding position of the adjacency matrix. However, if the value of cosine similarity fails to exceed the threshold value, then it is replaced with 0 in the adjacency matrix. The calculation is based on the following formula:

$$\cos(n_i, n_j) = \frac{n_i \cdot n_j}{|n_i|^2 + |n_j|^2} \quad (1)$$

$$A_{ij} = \begin{cases} \cos(n_i, n_j), & \text{if } i \neq j \text{ and } \cos(n_i, n_j) \geq w \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Here, n_i and n_j respectively correspond to the feature vectors associated with samples i and j . The parameter w reflects the average number of edges maintained per sample, including self-connections.

2.6. Graph transformer

The use of multiple candidate adjacency matrices by the graph transformer enables it to uncover new graph structures, thereby enhancing its ability to perform graph convolution and learn more robust node representations [43]. A soft selection approach is employed, wherein candidate adjacency matrices are generated using a 1×1 convolution layer C with parameter W_c , followed by a softmax function. The resultant adjacency matrices are then multiplied to obtain a new graph structure A , which is subsequently normalized using its degree matrix D .

$$Q = C(A; \text{softmax}(W_c)) \quad (3)$$

$$A = D^{-1} Q_1 Q_2 \quad (4)$$

2.7. Functional enrichment analysis

Functional Enrichment Analysis (FEA) is a bioinformatics technique utilized to investigate the functions of specific genes or proteins through identification of the cellular components, molecular functions, and biological processes associated with them, using their expression or sequence features. FEA can be employed as a tool for interpreting deep learning models. GO analysis and KEGG analysis were carried out for mRNA and methylation genes, utilizing the EnrichGO and EnrichKEGG functions of the R package “clusterProfiler”. For miRNA genes, GO analysis and KEGG pathway enrichment analysis were performed using the miRNA Enrichment and Annotation (miEAA) tool [44]. A p -value < 0.05 was considered statistically significant.

3. Results

3.1. Comparison of survival prediction results with other algorithms

Predictive models were constructed for six types of tumors, with hyperparameters optimized through the use of optuna [45] to achieve optimal prediction performance. To obtain a representative sample, the datasets were trained on 80% randomly selected samples, while retaining a consistent class ratio for each sample. The hyperparameters were identified via a 5-fold cross-validation process on the training samples, and the model's efficiency was evaluated using the remaining 20% holdout test set. Results were compared with those obtained using other algorithms, including Multimodal [46], GDP [47], Cox-nnet [39], DeepSurv [11], AECOX [48], DCAP [49], and DeepProg [50], with the aid of the C-index. Our models achieved superior predictive performance across all six types of tumors, as presented in Table 2. The predicted risk coefficients were used to categorized the samples into groups of high and low risk based on the median value. The resulting KM survival curves are presented in Fig. 2 and exhibit remarkable p -values ($p < 0.0001$).

3.2. Influence of combination of different omics data and hyperparameters on performance of survival prediction

To manifest the potential of integrating multiple omics data for improving survival predictions, we evaluated MFGAN's performance

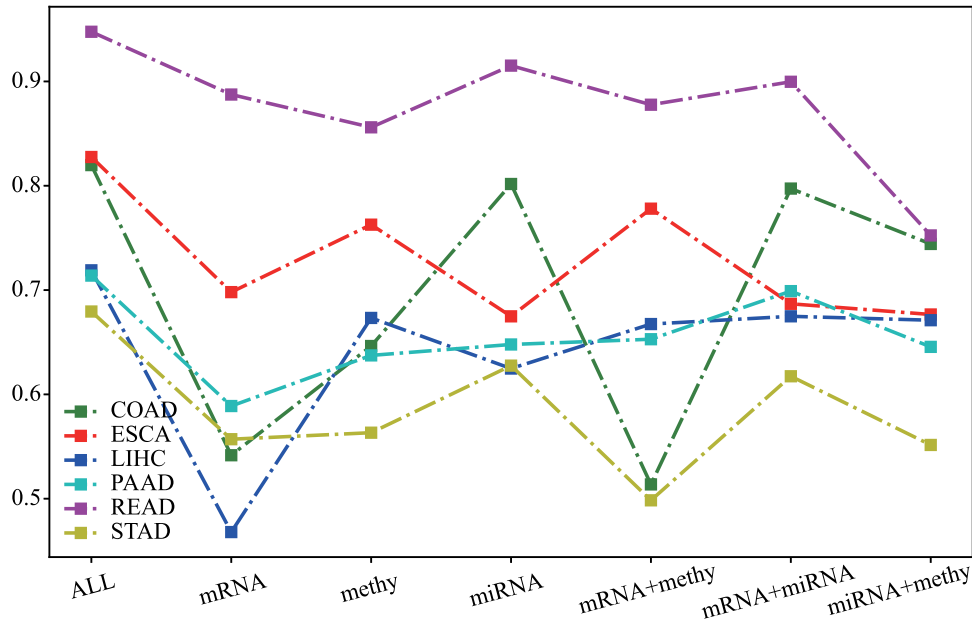


Fig. 3. Performance of MFGAN on survival prediction using different types of omics data.

Table 2

Our method MFGAN was compared to other algorithms using C-index across six types of tumors.

	MFGAN	Multimodal	GDP	Cox-nnet	DeepSurv	AECOX	DCAP	DeepProg
COAD	0.821	0.741	0.688	0.712	0.723	0.704	0.752	0.714
ESCA	0.828	0.782	0.714	0.722	0.753	0.698	0.794	0.683
LIHC	0.719	0.672	0.622	0.634	0.655	0.645	0.701	0.616
READ	0.948	0.883	0.772	0.798	0.804	0.832	0.904	0.835
PAAD	0.714	0.645	0.603	0.578	0.567	0.542	0.665	0.678
STAD	0.681	0.634	0.548	0.556	0.562	0.524	0.663	0.591

across all possible combination of omics data, including individual types of data (mRNA expression, DNA methylation, miRNA expression) and several combinations comprising mRNA expression + DNA methylation, mRNA expression + miRNA expression, DNA methylation + miRNA expression, and mRNA expression + DNA methylation + miRNA expression. The model trained with three types of omics data exhibited superior performance compared to models trained with single or two omics data types. However, using additional omics data does not invariably improve performance. For instance, the combination of DNA methylation and miRNA expression performed less favorably in all six tumor types than individual omics data. This is attributable to the distinct responses of diverse tumor types to specific omics data types, and some pairings of two omics data types can be disadvantageous. This emphasizes the significance of integrating multiple types of omics data and the effectiveness of MFGAN in achieving this. Fig. 3 presents the results in support of this claim.

To achieve optimum performance of MFGAN in survival prediction tasks, we optimized crucial hyperparameters such as the learning rate, optimization function, threshold w , GAN layer quantity, and number of hidden units within each layer. We applied OPTUNA with various sets of MFGAN hyperparameters during the tuning phase to enhance the mean AUC and AUPR scores in 5-fold cross-validation. Upon accomplishing the hyperparameter search, we employed the optimal parameters to construct the concluding model. Furthermore, we assessed the influence of distinct parameters on model performance and concluded that the chosen hyperparameters were all optimal, as presented in Fig. 4.

3.3. Identification and functional analysis of key genes in survival prediction

We conducted gene perturbation experiments to identify the top 30 genes with the most extensive impact on survival prediction models for each omics dataset. We have listed the important genes for survival prediction in STAD (see Table 3). To further investigate the function of these genes and interpret the mechanism of the model, we conducted functional enrichment analysis. We analyzed the functions of key genes in STAD and LIHC through literature analysis.

In STAD, the KEGG pathway analysis of mRNA and methylation key genes indicated significant enrichment of the Apelin signaling pathway. Through binding with its homologous receptor APJ, this pathway regulates various cellular processes [51]. Previous studies have reported a direct relationship observed between serum Apelin levels and C-reactive protein in GEC patients, suggesting that systemic inflammatory response might be linked to serum Apelin production [52]. Moreover, strong Apelin staining has been correlated with decreased overall survival in GC patients, when compared to low expression or weak Apelin staining [53]. The GO enrichment analysis disclosed that the structural component of the extracellular matrix, the extracellular matrix containing collagen, and regulation of blood circulation are linked to these genes as shown in Fig. 5A. In STAD, the miRNA key genes were majorly enriched in glycerophospholipid metabolism. Glycerophospholipids, also known as phospholipids, are critical molecules that regulate various cellular processes and contribute to defining cellular structure [54]. Throughout the entire growth process, cells participate notably in phospholipid metabolism. Furthermore, elevated

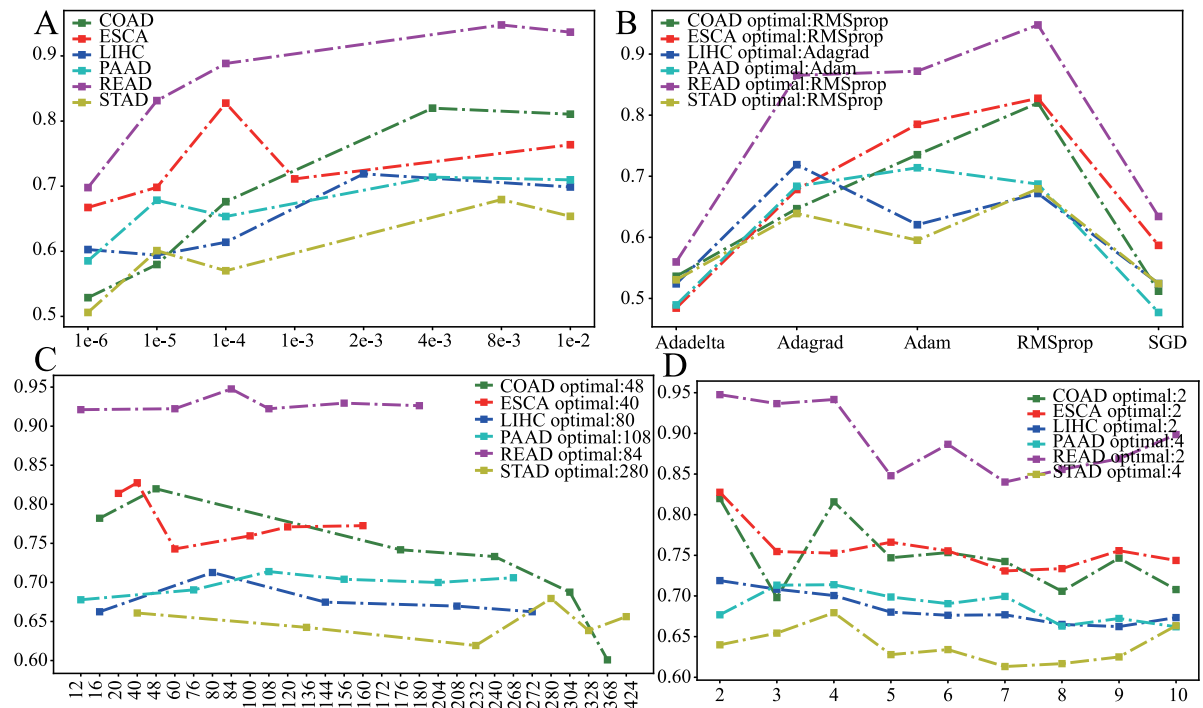


Fig. 4. The impact of hyperparameters on model performance. (A) Learning rate; (B) Optimization function; (C) Number of layer units; (D) Threshold W.

Table 3

Biomarkers in the STAD survival prediction model.

Biomarkers	
mRNA	MFAP4,MMP2,LBH,BASP1,CCN2,NT5E,AKR1B1,MYADM,TCIM,LTBP2,PLAT, SERPINE1,RGS2,PPP1R14A,RGS5,RAI14,GJA1,EDNRA,FZD7,PER1,EPDR1,COX4I2, B3GNT9,VCAN,DKK1,TPST1,LAMA4,PDPN,ANGPTL4,CCDC3
Methy	RGS22,NHLH2,POTEKP,CD38,AKR1B1,MYBPC1,CRYGA,TUBB8P4,GNB4, DPH6-AS1,SEMA3E,TNIP3,LINC01460,TCEAL3,ZNF331,ADRA1D,CYR61,DCP1B, GDF5,SLFN13,PPP1R3C,AURKC,CCL2,ZNF638,NMUR2,RNLS,DMRT2,IL12RB2, DACT2,LINC01077
miRNA	hsa-mir-4482,hsa-mir-653,hsa-mir-1293,hsa-mir-100,hsa-mir-6764,hsa-mir-7-1, hsa-mir-3143,hsa-mir-184,hsa-mir-15b,hsa-mir-4444-2,hsa-mir-3161,hsa-mir-4727, hsa-mir-9-3,hsa-mir-3133,hsa-mir-4664,hsa-mir-125b-2,hsa-mir-7-2,hsa-mir-3116-1, hsa-mir-125a,hsa-mir-1260b,hsa-mir-125b-1,hsa-mir-9-2,hsa-mir-218-2,hsa-mir-99b, hsa-let-7 g,hsa-mir-4793,hsa-mir-4739,hsa-mir-7703,hsa-mir-223,hsa-mir-6511b-2

nucleoside levels, particularly adenosine, may result in shortened survival in GC patients [55]. Insulin-activated receptor activity, insulin receptor complex, and transcytosis were among the functions revealed by the GO enrichment analysis as shown in Fig. 5B.

The KEGG pathway analysis of mRNA and methylation key genes in LIHC revealed major enrichment in the p53 signaling pathway. This pathway regulates critical cellular processes like DNA repair, cell cycle, apoptosis, and differentiation via DNA-protein and protein-protein interactions [56]. Low expression of p53 has been linked with poorer prognosis in gastric cancer [57]. The GO function enrichment was observed in virus receptor activity, catalytic step 1 spliceosome, and viral process, show in Fig. 5C. The miRNA key genes analysis in KEGG pathways identified significant enrichment in Proteasome, an essential compartmentalized protease present across all archaea and eukaryotes. Various studies have shown that proteasome activators not only regulate proteasome function but are also associated with different malignant tumors, thereby playing a role in prognosis prediction [58], including gastric cancer [59]. The GO function enrichment was observed in sequence-specific mRNA binding, proteasome regulatory particle, and downregulation of stress-activated signaling via MAPK cascade, show in Fig. 5D.

3.4. Comparison of drug response prediction by MFGAN with other methods and impact of different omics data

We conducted 5-fold cross-validation using AUC and AUPR to evaluate the performances of MFGAN and other models (DeepCDR [28], GraphCDR [31], Random Forest [60], RefDNN [61], tCNNs [26], NRL2DRP [62]) on the GDSC dataset. Considering that the proportion of sensitive samples (1240) to resistant samples (12534) is over 1:10, we conducted three experiments on all algorithms, including: raw data, data oversampling, and data undersampling. MFGAN outperformed all the compared algorithms, and its AUC scores in the three experiments respectively exceeded the two best algorithms DeepCDR and GraphCDR: 2.12% and 1.19% for raw data, 4.55% and 3.38% for oversampled data, and 2.25% and 1.69% for undersampled data. The corresponding AUPR scores exceeded by MFGAN were 3.29% and 1.30% for raw data, 6.16% and 2.45% for oversampled data, and 2.79% and 1.68% for undersampled data. We further evaluated the impact of combining different omics data on the performance of MFGAN, including three single omics data and three combinations of two omics data. It can be seen that the performance of these data combinations is worse than that of the model using three omics data, whether in

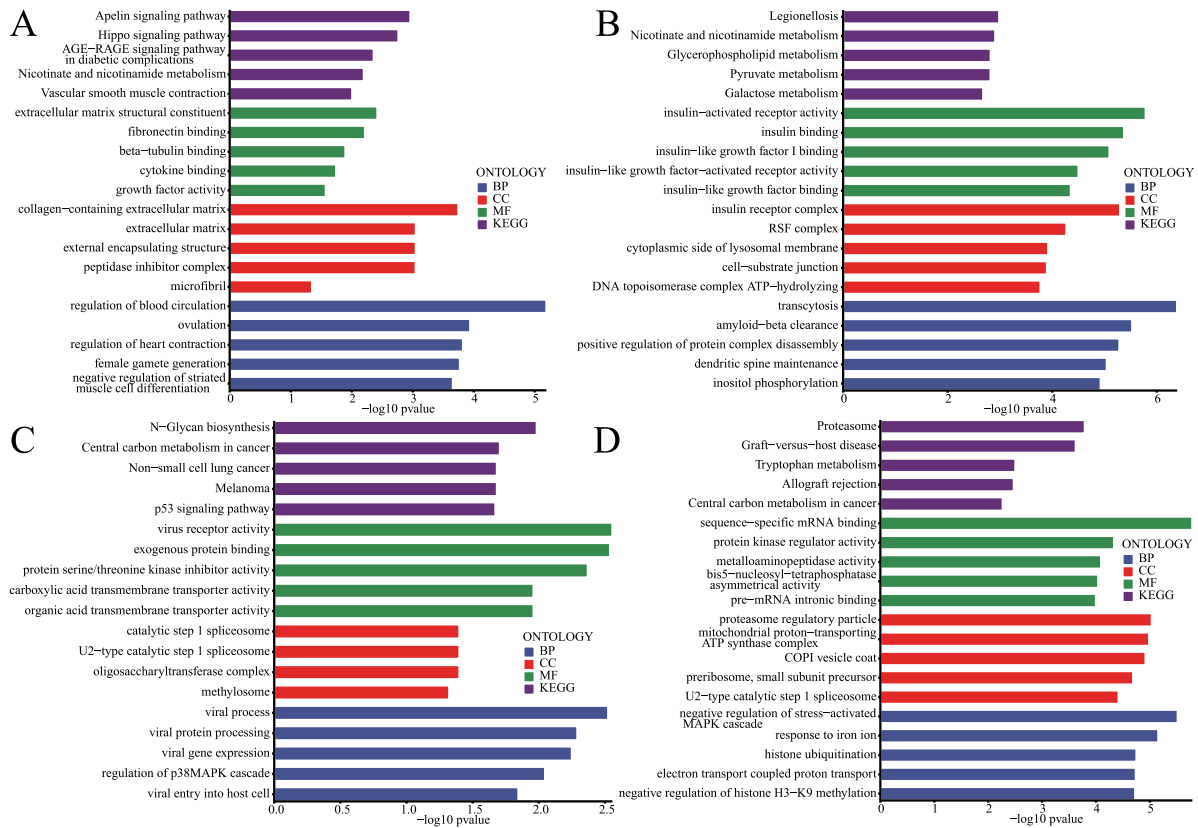


Fig. 5. Functional enrichment analysis of key genes for survival prediction models in STAD and LIHC. (A) Functional enrichment analysis of mRNA expression and DNA methylation expression in STAD. (B) Functional enrichment analysis of miRNA expression in STAD. (C) Functional enrichment analysis of mRNA expression and DNA methylation expression in LIHC. (D) Functional enrichment analysis of miRNA expression in LIHC.

Table 4

Comparison of MFGAN with other methods in drug response prediction and the impact of different omics data combinations on MFGAN.

	Raw		Oversampling		Undersampling	
	AUC	AUPR	AUC	AUPR	AUC	AUPR
DeepCDR	0.7151	0.1903	0.8371	0.7911	0.7397	0.7258
GraphCDR	0.7244	0.2102	0.8488	0.8282	0.7453	0.7369
Random Forest	0.4994	0.0545	0.6971	0.7122	0.5987	0.7025
RefDNN	0.6011	0.1187	0.7797	0.7816	0.6065	0.6431
tCNNs	0.6215	0.1325	0.7958	0.7983	0.7021	0.6959
NRL2DRP	0.6628	0.1528	0.7688	0.7882	0.6782	0.6821
MFGAN	0.7363	0.2232	0.8826	0.8527	0.7622	0.7537
MFGAN_mRNA	0.7209	0.2004	0.8119	0.7771	0.7124	0.7266
MFGAN_methy	0.7111	0.2015	0.8106	0.7939	0.7233	0.7251
MFGAN_CNV	0.7243	0.2068	0.8019	0.7881	0.7366	0.7345
MFGAN_mRNA_methy	0.7097	0.2116	0.8622	0.8353	0.7476	0.7428
MFGAN_CNV_methy	0.7217	0.2171	0.8748	0.8489	0.7387	0.7221
MFGAN_mRNA_CNV	0.7309	0.2155	0.8254	0.8095	0.7511	0.7408

raw data, oversampled data, or undersampled data. However, it does not necessarily mean that the performance of two omics data is better than that of one omics data, which may be because different omics data affect each other, and this influence may be beneficial, harmful, or even worse. All the results are shown in Table 4.

3.5. The influence of hyperparameters on performance and tumor-specific experiments

In order to achieve the best performance of MFGAN in drug response prediction tasks, we used the OPTUNA algorithm to optimize important hyperparameters such as learning rate, optimization function, threshold w , number of GAN layers, and the number of neurons in the hidden

layer in each layer, similar to the survival prediction model. The optimal parameters obtained from the search were used to construct models for the original data, data oversampling, and data undersampling. We evaluated the effects of different values of each hyperparameter on model performance and demonstrated that the hyperparameters we used were optimal. The results of the model using data oversampling are shown in Fig. 6.

To evaluate the tumor specificity of MFGAN, we assessed the performance of MFGAN and other algorithms on five tumor datasets according to the tumor type to which the cell lines belong. As shown in Fig. 7A, COAD and READ were considered as a disease based on the cell line annotation file, which is related to their similarity in pathology and clinical aspects, resulting in the highest number of CDRs, accounting for 34% of all CDRs, while the CDRs of the other four types of tumors were relatively evenly distributed. As shown in Fig. 7B, MFGAN performed better than the other six methods. For example, the AUC values of each tumor dataset were 0.8822, 0.8726, 0.9241, 0.9067, and 0.8742, while the AUC values of GraphCDR, the best-performing method among the six comparison methods, were 0.8523, 0.8496, 0.9079, 0.8886, and 0.8422, respectively. The experimental results can demonstrate that our model can accurately attain consistent exceptional performance concerning targeted tumor therapies across different types of tumors.

3.6. Prediction of unknown CDRs and functional analysis of key genes

While previous experiments have been conducted using known CDRs, the prediction of unknown CDRs is of significant importance in advancing cancer drug therapy. To achieve this, we trained the MFGAN model using all known CDRs and utilized it to predict new responses (i.e., unknown CDRs). To eliminate the randomness in predictions and avoid overfitting to the available data, we conducted five training runs and took the average of the results as the final

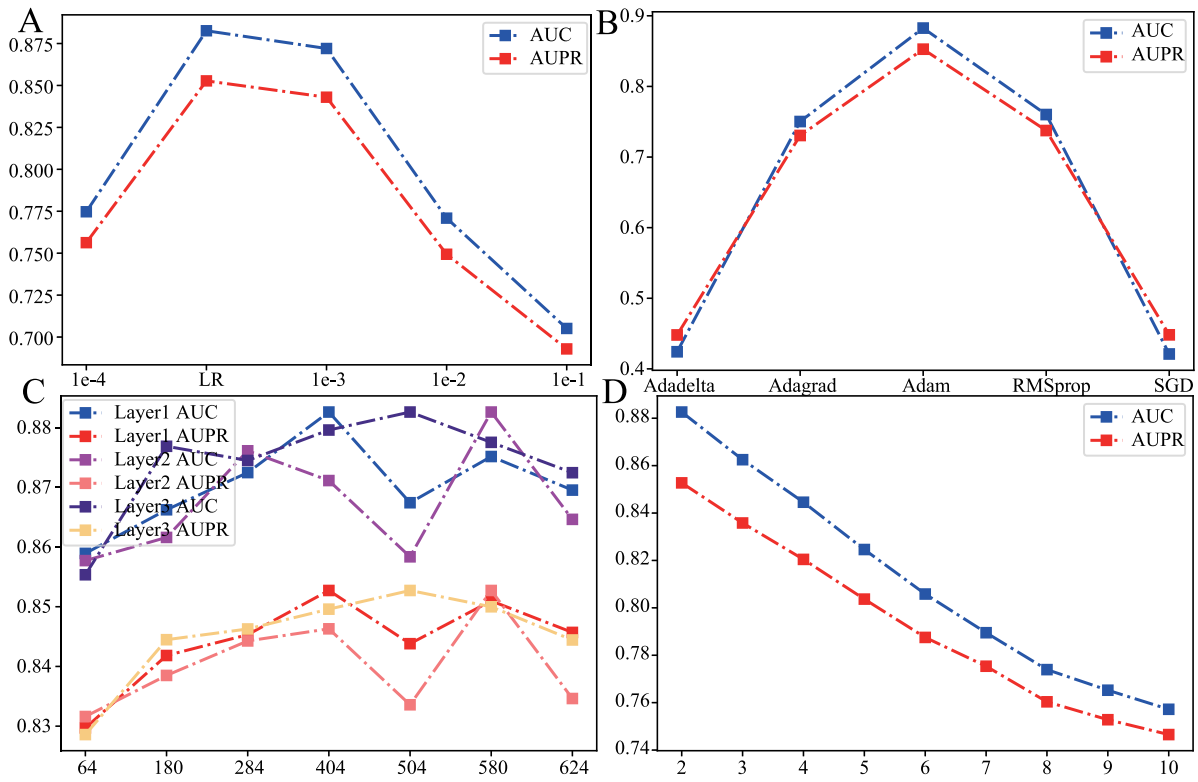


Fig. 6. The impact of hyperparameters on data oversampling model performance. (A) Learning rate; (B) Optimization function; (C) Number of layer units; (D) Threshold W.

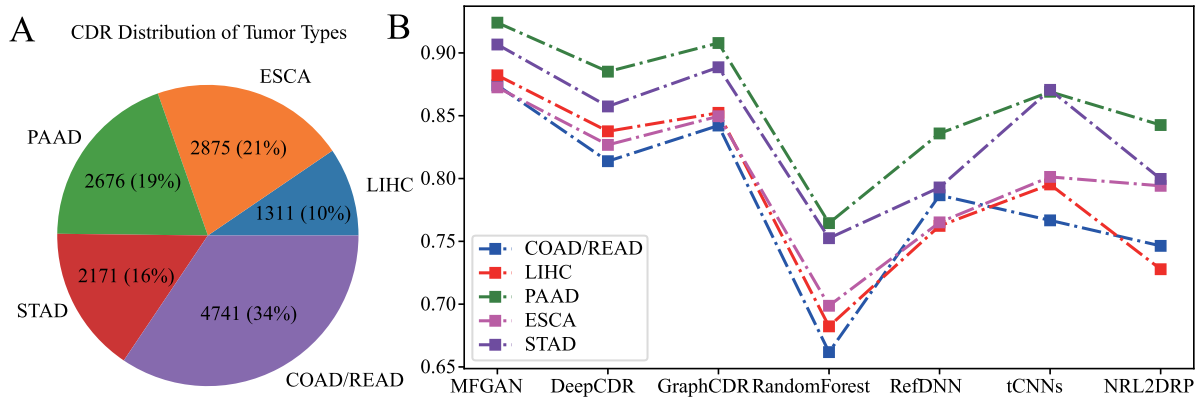


Fig. 7. MFGAN's tumor specificity experiment on data oversampling. (A) The distribution of CDR in 5 types of tumors, with COAD/READ, ESCA, PAAD, STAD, and LIHC accounting for 34% (4741), 21% (2875), 19% (2676), 16% (2171), and 10% (1311) of all CDRs, respectively; (B) AUC values of MFGAN, DeepCDR, GraphCDR, Random Forest, RefDNN, tCNNs, and NRL2DRP model.

prediction. The prediction results are shown in Table 5. The mTOR signaling pathway plays a crucial function in tumor development and is overexpressed or constitutively upregulated in a multitude of malignant tumors. Yang et al. have suggested rapamycin as a potential treatment approach for colon cancer treatment [63]. Additionally, AZ628, a RAF kinase inhibitor, blocks CRAF activation by linkage with the ATP binding region of RAF kinase. Peter et al.'s research found that downregulation of RKIP inhibition may reduce the ability of esophageal cancer to establish disseminated diseases, thus providing potential therapeutic significance [64]. Quizartinib is a reversible inhibitor of serine/threonine-protein kinase, which mainly affects the MAPK pathway by acting on the MEK protein and inhibiting cell proliferation. Through their research, Meng et al. have found that HGF/c-MET-ERK1/2 signaling pathway upregulates the expression of ETV4. Moreover, combined therapy with a PD-L1 inhibitor and Trameetinib, a MAPK pathway blocker has been found effective in improving

the immune-suppressed microenvironment, offering a promising therapeutic approach for liver cancer treatment [65]. Kei et al. showed that the tumor regression caused by quizartinib demonstrates its potential efficacy for pancreatic cancer patients [66]. PHA-665752 is a c-Met kinase inhibitor and an ATP competitive active site inhibitor. Chen et al.'s research showed that PHA-665752 inhibits the growth of GC cells by inhibiting MET phosphorylation and MAPK and AKT signal transduction [67].

To functionally analyze the omics data for drug response prediction, we identified the top 30 genes from each omics data that have significant impacts on drug response prediction. In order to determine the extent of enrichment of specific biological processes or pathways in the set of genes, we conducted functional enrichment analysis. In Table 6, we present the key genes along with the obtained results of the enrichment analysis. Remarkably, the KEGG pathway analysis of the key genes demonstrated a marked enrichment in the transcriptional

Table 5
The highest predicted probability of unknown CDR in each tumor.

Drug (Pubchem)	Drug name	Cell line (depMapID)	Cell line name	Tumor type	Probability
5384616	Rapamycin	ACH-000236	SW1417	COAD/READ	0.9615
11676786	AZ628	ACH-000823	KYSE-140	ESCA	0.9716
11707110	Trametinib	ACH-000422	SNU-475	LIHC	0.9668
11707110	Trametinib	ACH-000270	HPAC	PAAD	0.9745
10461815	PHA-665752	ACH-000225	ECC12	STAD	0.9757

Table 6
Identification of key genes and functional enrichment analysis for drug response prediction.

CNV	MYCL,DNM2,EPHA3,NFKB2,MYC,EXT1,TLX1,CSF3R,FOX P1,MITF,B2M,CASP3,ETV1,FHIT,TP53,CPEB3,PTEN,FANCF,SIRPA,CALR,IRF4,PDGFB,ZEB1,MYO5A,RPN1,ATR,ARAF,MLLT10,MACC1,MB21D2
mRNA	DNMT3A,BCORL1,POLD1,JAK2,PRF1,HOXA9,ARHGEF10L,MECOM,MYH11,SALL4,SLC34A2,USP44,MEN1,COL2A1,CD79B,TFE3,TCF12,FLI1,FGFR1,NRG1,SEPT9,FOX L2,LMO1,BCL6,DCC,ETV4,PAX7,PDE4DIP,GATA2,SLC34A2,
methy	CDKN2A,FOXO3,SDHAF2,SOC S1,SPECC1,LRIG3,BCLAF1,H3F3B,MUC4,SDHC,DNMT3A,GNAS,NFATC2,CASP8,MSH2,CTNND1,N4BP2,HOXC13,KIF5B,SIRPA,STAT6,APC,NRG1,ERBB3,MYO5A,PCDC1LG2,LATS2,SETD1B,ATP1A1,PCBP1,
KEGG	Transcriptional misregulation in cancer, Melanoma, EGFR tyrosine kinase inhibitor resistance, MicroRNAs in cancer, Colorectal cancer
BP	myeloid cell differentiation, leukocyte differentiation, negative regulation of cyclin-dependent protein serine/threonine kinase activity, negative regulation of cyclin-dependent protein kinase activity, regulation of hemopoiesis
CC	transcription repressor complex, endosome lumen, RNA polymerase II transcription repressor complex, cytolytic granule, plasma membrane protein complex
MF	DNA-binding transcription activator activity, RNA polymerase II-specific,DNA-binding transcription activator activity, chromatin binding, E-box binding, DNA-binding transcription factor binding

misregulation in cancer pathway. It is noteworthy that the transcriptional misregulation is a defining hallmark of cancer and the drugs targeting intracellular signaling pathways have substantially enhanced the progression-free survival rates of previously untreatable cancer patients. Additionally, the dysregulation of key transcription factors not only characterizes the cancer phenotype. Furthermore, it plays a central role in both the development and maintenance of the disease [68]. The GO functional analysis showed enrichment in myeloid cell differentiation, transcription repressor complex, and DNA-binding transcription activator activity.

4. Discussion and conclusion

Several prior computational approaches have employed a single omic data type for the purpose of estimation survival and drug response. However, as a single omic type presents only a limited view of cancer, the performance of these methods is restricted. Multi-omic data analysis can yield further details on cancer and improve prediction performance. We introduce a novel multi-functional deep learning model integrated with multi-omic data for survival and drug response prediction in DST. Our model exhibits superior performance in comparison to other advanced algorithms, demonstrating its superiority and multi-functionality. Through an ablation experiment, we show the

model’s powerful ability to integrate multi-omic data. Extraction of key genes from omic data and their functional enrichment analysis reveal their association with the survival prognosis and drug therapy of DST, providing some explanation for the high performance of our deep learning model, and resolving the black box problem of deep learning models. Predictions for unknown CDR can offer constructive suggestions for DST treatment.

Despite the proven effectiveness and reliability of our method in predicting survival and drug response, there are still several issues that require further discussion. Firstly, we have found that the C-index values for predicting survival for different cancers vary significantly. One possible reason for this could be the oversight of important clinical factors such as acknowledged tumor purity in TCGA. Therefore, we recommend including some essential clinical data as input in subsequent studies. Secondly, although we have used various methods for selecting features to address the challenge of limited samples and a large number of features, it is still possible to overlook some significant features. Thus, we plan to combine the feature selection component with the model to form a more comprehensive model. Thirdly, despite using three types of omic data in this study, we recognize that there are still limitations. Therefore, we intend to include additional types of omic data, such as pathological image data and copy number variation data, in future work. Finally, despite utilizing a public database covering multi-omics data, there are still limitations in the scale of the dataset. There are plans to incorporate data from databases such as ICGC and GTEx to expand the dataset, providing a more comprehensive reflection of real-world samples and improving the model’s generalization capability. In conclusion, we developed a reliable and efficient multi-omics integrated model, referred to as MFGAN, for predicting survival and drug response in DST, creating new opportunities for in-depth analysis and deep learning-assisted cancer survival prognosis, and exploration of drug combinations in the future. This may aid in the clinical diagnosis of DST and enable prognosis guidance based on omics data in the near future. Novel and consistent drug-cancer associations can be identified through newly predicted drug response, which can aid in drug repositioning. Therefore, this has the potential to advance the field of precision medicine and contribute to the rapidly developing field of oncology.

Declaration of competing interest

The authors declare that they have no competing interests.

Data availability

The data that has been used is not confidential.

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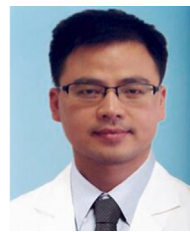
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